

ValuationLab: Automated DCF & Monte Carlo Platform

Comprehensive Technical Synopsis, Mathematical Formulations, and Financial Logic

Framework: FastAPI (Python) • React 18 • Plotly.js

Quant Models: Linear Fade DCF • 7-Var Cholesky Monte Carlo • Student's t

Markets: United States (NYSE/NASDAQ) & India (NSE/BSE) **AI & Export:** Groq LLaMA 3.1 70B • 6-Sheet OpenPyXL Model

1 Executive Overview & Architecture

ValuationLab is an institutional-grade equity valuation platform engineered to automate fundamental financial modeling and stochastic risk evaluation. Traditional spreadsheet DCF models rely on brittle point estimates for revenue growth, operating margins, and discount rates, suffering from false precision. ValuationLab solves this by combining automated statement extraction with rigorous quantitative econometrics:

1. Automated Ingestion

Pulls 3-statement financials via yfinance and live 10-year risk-free rates via FRED API. Automatically aligns historical dates across statements.

2. Stochastic Modeling

Simulates 10,000 iterations using Cholesky decomposition across 7 correlated risk variables with fat-tailed Student's t-distributions and reject-and-resample filtering.

3. Institutional Reporting

Generates AI-written equity research notes via Groq LLaMA 3.1 70B and exports audit-ready 6-sheet financial model workbooks in Excel (.xlsx).

DATA INGESTION

yfinance • FRED API
Live Macro • Dual-Market

DATA LAYER & RATIOS

3-Stmt Date Intersection
Non-Cash NWC • Δ NWC
CapEx % • D&A % • CAGR

QUANT ENGINES

CAPM & Derived WACC
Linear Growth Fade DCF
Cholesky Monte Carlo (10k)

INTERFACE & EXPORT

React 18 / Plotly UI
Groq LLaMA 3.1 70B
openpyxl 6-Tab Model

2 Dual-Market Jurisdictional Profiles

ATTRIBUTE	UNITED STATES (US)	INDIA (IN)	MECHANISM IN CODE
Market Detection	Default / Standard tickers (AAPL, NVDA)	Tickers ending in .NS or .B0	market_config.detect_market()
Risk-Free Rate (R _f)	FRED Series DGS10 (10Y Treasury Yield)	FRED Series INDIRLT01STM (10Y G-Sec Yield)	get_live_risk_free_rate() with 1-hr cache
Fallback Rate (R _f)	4.20%	7.10%	Automatic fallback if API key absent or rate limit hit
Market Return (R _m)	10.00% (S&P 500 Historical Benchmark)	13.00% (Nifty 50 Historical Benchmark)	Computes Equity Risk Premium (ERP = R _m - R _f)
Corporate Tax Rate (T)	21.00% (Federal Statutory Corporate Tax)	25.17% (Section 115BAA incl. Surcharge & Cess)	Applied to EBIT for NOPAT and debt tax-shield
Currency & Scales	USD (\$), Billions (B), Millions (M)	INR (₹), Crores (Cr), Lakhs (L)	Dynamic UI and Excel number formatting strings

3 Three-Statement Cleaning & Feature Engineering

The raw data layer in `engine/data_layer.py` handles alignment, alternate column mappings, and normalization:

- **Temporal Alignment:** Transposes financial statements and intersects dates: `dates = IncStmt.index ∩ BalSheet.index ∩ CashFlow.index`, guaranteeing identical accounting periods across all line items.
- **Indian Reporting Taxonomy:** Resolves alternate line items (e.g. *"Total Revenue From Operations"* → *"Total Revenue"*, *"Current Borrowings"* → *"Current Debt"*).
- **Non-Cash Working Capital:** Removes cash and short-term interest-bearing debt to prevent double-counting:

NET WORKING CAPITAL & REINVESTMENT INTENSITY EQUATIONS

```
NWC_t = (Current_Assets_t - Cash_t) - (Current_Liabilities_t - Short_Term_Debt_t)
ΔNWC_t = NWC_t - NWC_{t-1}
CapEx_Pct = Mean( |CapEx_t| / Revenue_t ),   D&A_Pct = Mean( D&A_t / Revenue_t ),   ΔNWC_Pct = Mean(
ΔNWC_t / Revenue_t )
```

4 Cost of Capital (WACC) & Multi-Stage DCF

CAPM & Capital Structure

WACC is computed with market-value equity and book-value debt:

WACC FORMULATION

```
K_e = R_f + β × (R_m - R_f)
K_d = |Interest_Expense / Total_Debt|   [fallback: 4%]
W_e = Mkt_Cap / (Mkt_Cap + Total_Debt),   W_d = 1 - W_e
WACC = (W_e × K_e) + [W_d × K_d × (1 - T)]
```

Linear Growth Fade Logic

Growth decays smoothly to terminal GDP growth ($g = 2.5\%$):

FORECAST SCHEDULE

```
growth_schedule = np.linspace(g_start, g_terminal, 5)
Revenue_{t} = Revenue_{t-1} × (1 + growth_schedule[t])
NOPAT_t = Revenue_t × EBIT_Margin × (1 - T)
UFCF_t = NOPAT_t + (Rev_t × D&A%) - (Rev_t × CapEx%) - (Rev_t × ΔNWC%)
```

Terminal Value & Enterprise-to-Equity Valuation Bridge

```
Terminal_Value (TV) = [ UFCF_5 × (1 + g_terminal) ] / ( WACC - g_terminal )
PV(TV) = TV / (1 + WACC)^5,   PV(Explicit) = ∑ [ UFCF_t / (1 + WACC)^t ]
Enterprise_Value (EV) = PV(Explicit) + PV(TV)
Equity_Value = max( 0.0, EV + Cash_and_Equivalents - Total_Debt )
Implied_Share_Price = Equity_Value / Shares_Outstanding
```

5 Cholesky-Correlated Fat-Tailed Monte Carlo Engine

The Monte Carlo engine (engine/monte_carlo.py) conducts 10,000 simulations using quantitative econometrics rather than naive unlinked random draws:

The 7 Correlated Risk Variables

- Revenue Growth (g):** Mean = g_hist, Std = max(σ_g , 2%)
- EBIT Margin (m):** Mean = m_hist, Std = max(σ_m , 1.5%)
- Risk-Free Rate (R_f):** Mean = Live FRED rate, Std = 100 bps
- Equity Beta (β):** Mean = Reported Beta, Std = 0.15 error
- Risk Premium (ERP):** Mean = $R_m - R_f$, Std = 150 bps market vol
- Cost of Debt (K_d):** Mean = Base K_d, Std = 100 bps spread shift
- Equity Weight (W_e):** Mean = Base W_e, Std = 5% capital drift

Mathematical Formulation

CHOLESKY & ECONOMIC WACC
 $\Sigma_{\{7 \times 7\}}$: empirical corr($\rho_{\{g,m\}}$) between growth & margin
 $\Sigma = L \times L^T$ (Cholesky factor L)
 $Z \sim \text{Student}_t(df=5)$ [3x oversampling for fat tails]
 $X = \mu + (L @ Z) \times \sigma$

WACC derived economically per iteration:
 $K_{\{e,i\}} = R_{\{f,i\}} + \beta_i \times \text{ERP}_i$
 $\text{WACC}_i = (W_{\{e,i\}} \times K_{\{e,i\}}) + [(1 - W_{\{e,i\}}) \times K_{\{d,i\}} \times (1 - T)]$

Reject-and-Resample Filter

Draws are discarded and resampled if: (1) $\text{WACC}_i \leq g_{\text{terminal}} + 0.5\%$ (preventing asymptotic division by zero in Gordon Growth), (2) $\text{Margin}_i \leq 0$, (3) $K_{\{d,i\}} \leq 0$, or (4) $R_{\{f,i\}} \leq 0$. No artificial clamping is applied.

6 Valuation Cross-Check & Statistical Verdict

EV/EBITDA Multiple Cross-Check

Validates whether DCF intrinsic value deviates unrealistically from current market multiples:

```
EBITDA = Trailing_EBIT + Trailing_D&A
Market_EV_EBITDA = (Mkt_Cap + Debt - Cash) / EBITDA
Model_EV_EBITDA = DCF_Enterprise_Value / EBITDA
Multiple_Gap = |Model - Market| / Market * 100
Status = PASS if Gap < 50% else WARNING
```

Percentile-Based Verdict Matrix

Investment verdicts are tied directly to the Monte Carlo probability distribution:

SIGNAL	CONDITION	ACTIONABLE MEANING
STRONG BUY	Price < P5	Cheaper than 95% of simulated outcomes
BUY	Price < P25	Trades below 25th percentile downside
HOLD	$P25 \leq \text{Price} \leq P75$	Within interquartile fair value range
SELL	Price > P75	Trades above 75th percentile outcome
STRONG SELL	Price > P95	Requires extreme, low-probability bull case

7 AI Equity Research & Multi-Sheet Excel Export

Groq LLaMA 3.1 70B AI Research Note

The AI generator (`ai/summary_generator.py`) sends the complete quantitative output payload to Groq's high-throughput LLaMA 3.1 70B model:

- **Strict JSON Contract:** Enforces a 10-section JSON schema, eliminating unstructured markdown parsing errors.
- **Institutional Synthesis:** Produces executive summaries, market implied growth expectations, margin drivers, risk matrices, and valuation commentary.
- **UI Card Rendering:** Directly rendered into modular cards by `AIRenderer.jsx` with client-side PDF export.

OpenPyXL 6-Sheet Audit Workbook

The workbook generator (`export/excel_export.py`) compiles an institutional financial model workbook:

- **Sheet 1: Cover:** Overview, current price, DCF & MC values, verdict badge.
- **Sheet 2: DCF Model:** Complete forecast table, UFCF build, Gordon Growth TV, EV bridge.
- **Sheet 3: WACC:** Detailed CAPM breakdown, cost of debt, and capital weights.
- **Sheet 4: Monte Carlo:** Summary statistics, P5–P95 percentiles, draw parameters.
- **Sheet 5: Sensitivity:** Formatted 2D data tables with conditional formatting heatmaps.
- **Sheet 6: Assumptions:** Complete audit trail of raw inputs and active overrides.

8 Frontend Views & REST API Endpoints

Frontend User Interface Tabs

- **Overview:** Intrinsic value cards, verdict badge, EV/EBITDA status, AI memo, PDF download.
- **DCF Model:** 5-year explicit projection table & EV-to-Equity waterfall reconciliation.
- **Monte Carlo:** Interactive Plotly histogram with P5/P95 confidence interval & median line.
- **Sensitivity:** Two 2D heatmaps: (1) Growth vs WACC and (2) Margin vs WACC.
- **Trends:** Revenue CAGR bar chart, margin expansion slope, and proxy Free Cash Flow.
- **Sidebar:** Real-time Assumption Sliders for instant scenario re-simulation.

REST API Reference (Port 8000)

ROUTE	TYPE	DESCRIPTION
/api/health	GET	Liveness probe
/api/macro-rates	GET	Live FRED risk-free rates & profile (US/IN)
/api/search	GET	Yahoo Finance ticker auto-complete proxy
/api/analyze	POST	Full valuation & Monte Carlo pipeline
/api/ai-summary	POST	Groq LLaMA 3.1 70B research report generation
/api/export/excel	POST	Binary stream of 6-sheet .xlsx workbook